Assignment 2

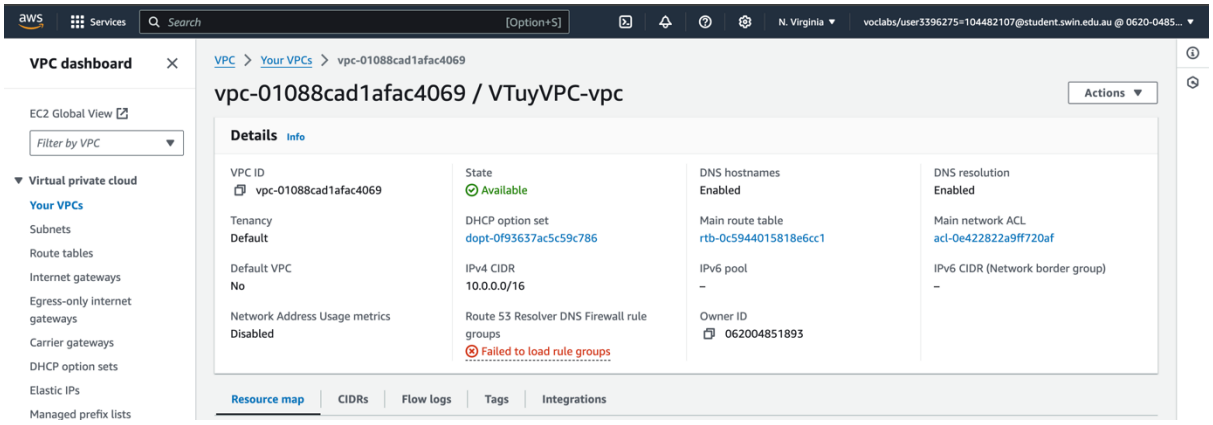
Name: Van Praseth Tuy
ID: 104482107
Tutor Time: Friday 4:30 PM – 6:30 PM
Date of Submission: 6th October 2024

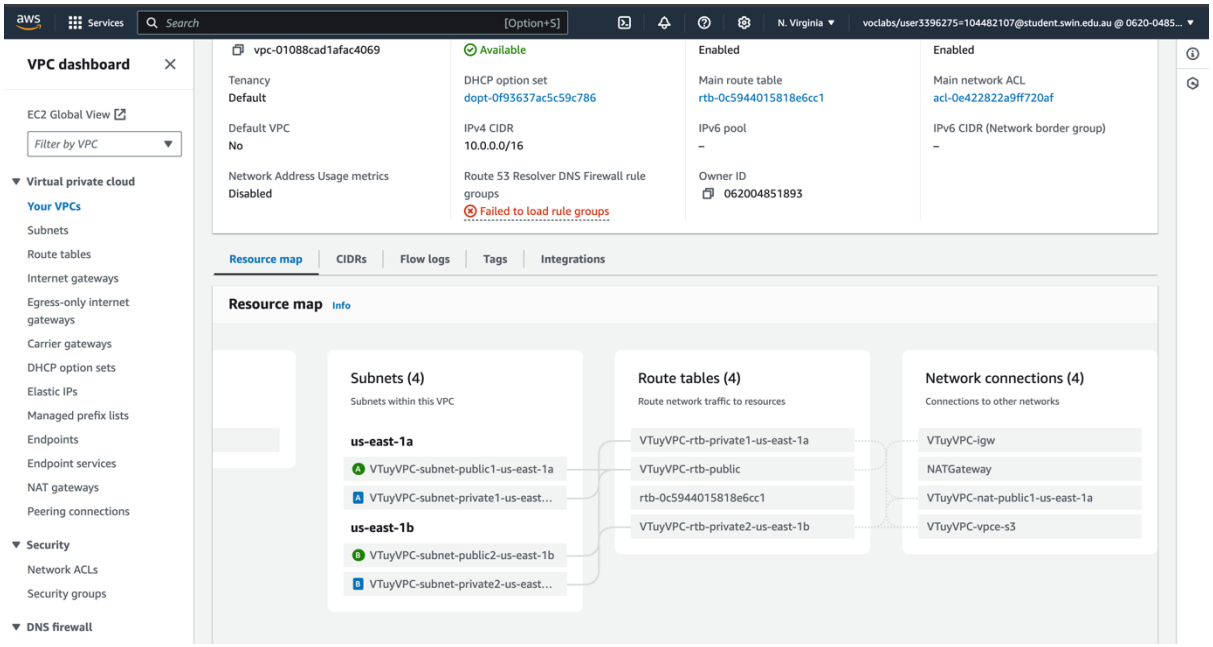
I. INTRODUCTION

This assigment is to deploy a website to upload the photo to the database by the auto-scaling group which is created by the load balancer.

2.1 - VPC

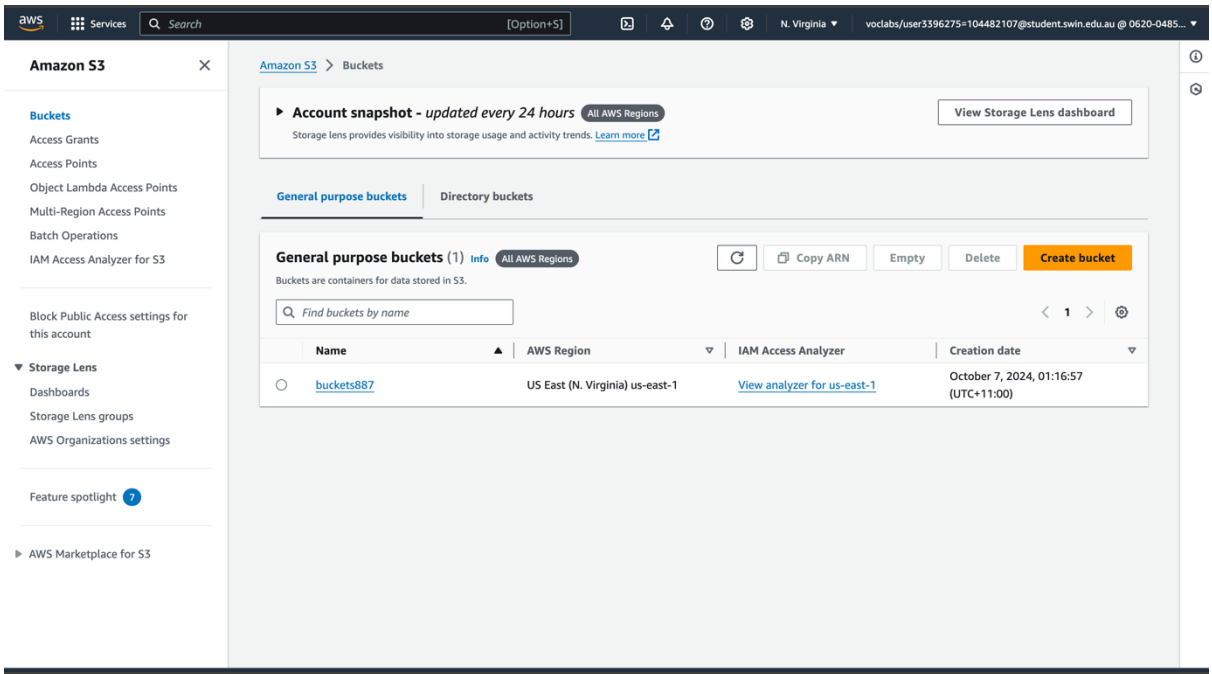
I have created VPC by following the step in a Assignment 1b setting the VPC name VTuyVPC.

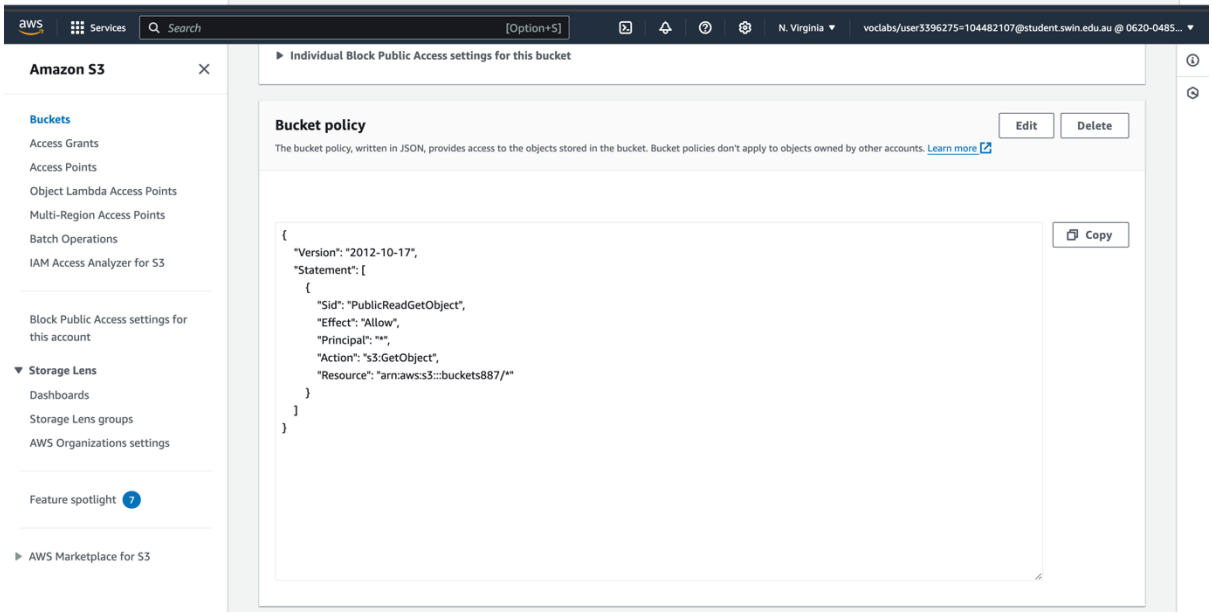


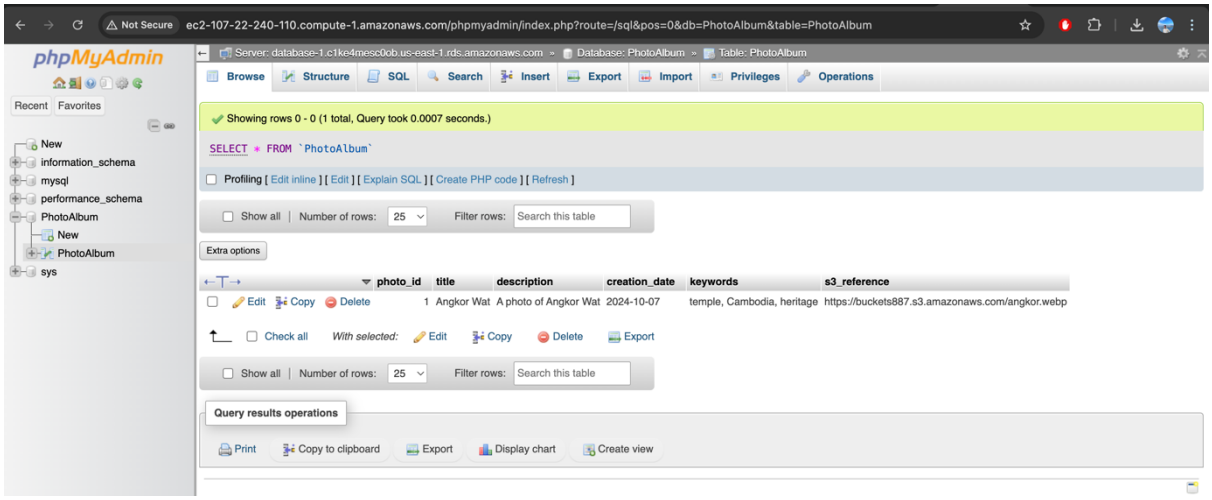


2.2 – S3 photo storage

After that, I have created S3 bucket to store and upload the image as the database and puttingf the bucket policy and ceate table on MySQL through public DNS of Dev server instance.

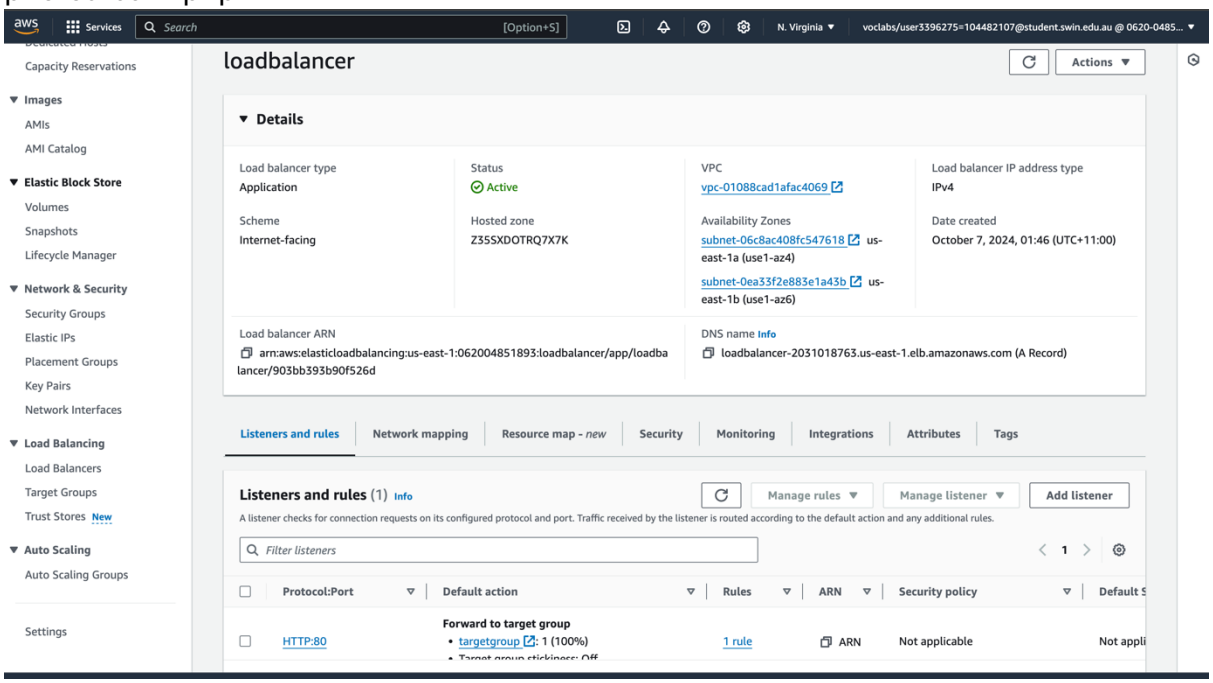






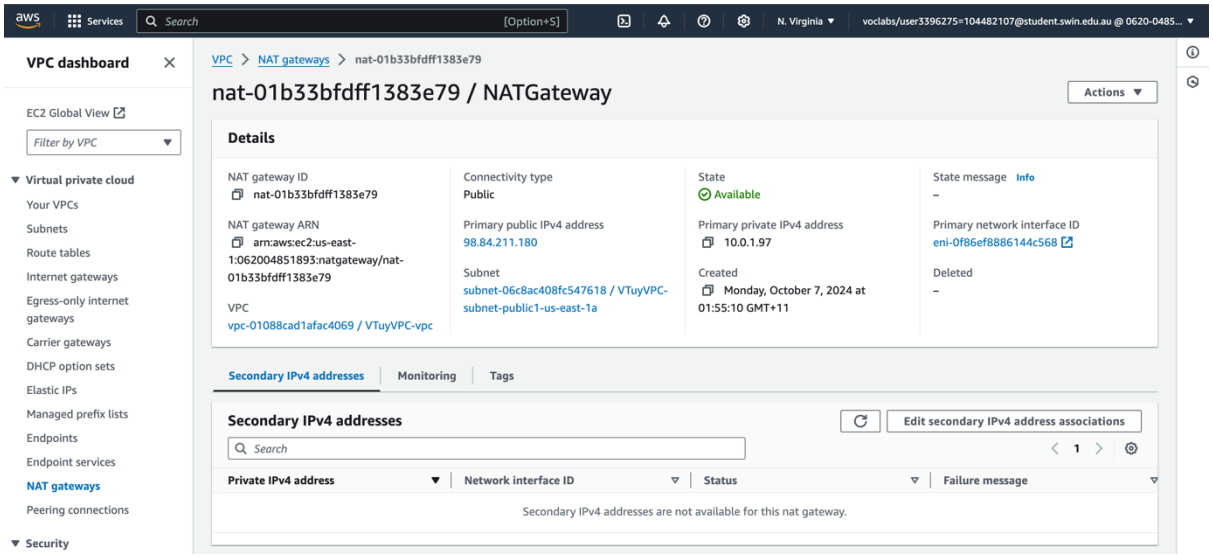
2.3 – Load balancing

Then, I have created load balancing to distribute traffic across instances, with a security group for protection and it will monitor instance health by checking the photoalbum.php.



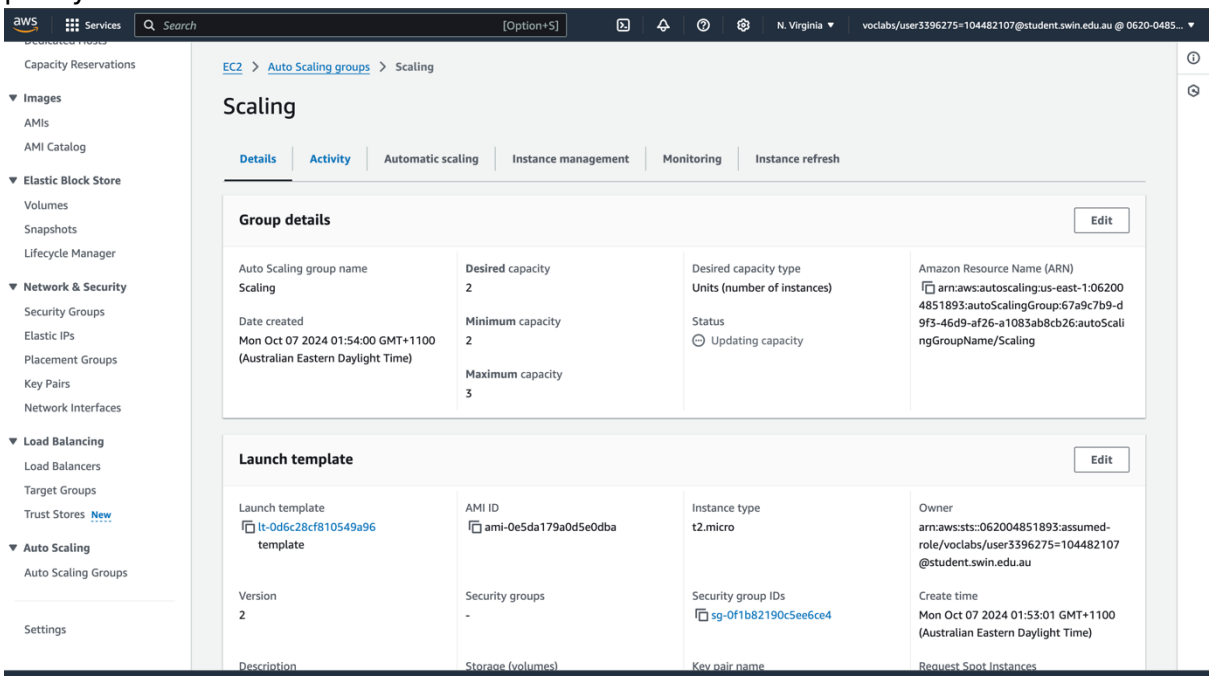
2.4 – Network Address Translation (NAT)

Moreover, I have created native address translation to allow local network to communicate to internet.



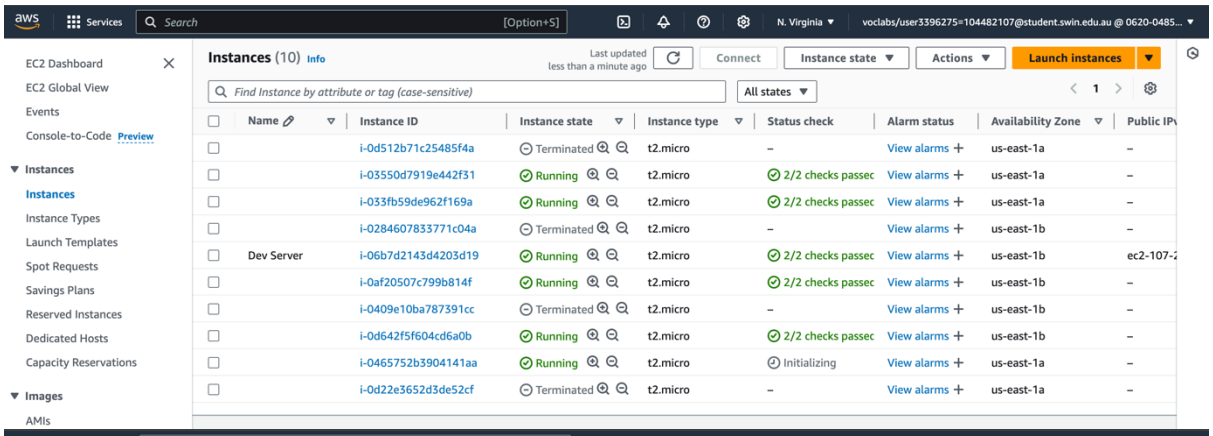
2.5 – Auto scaling

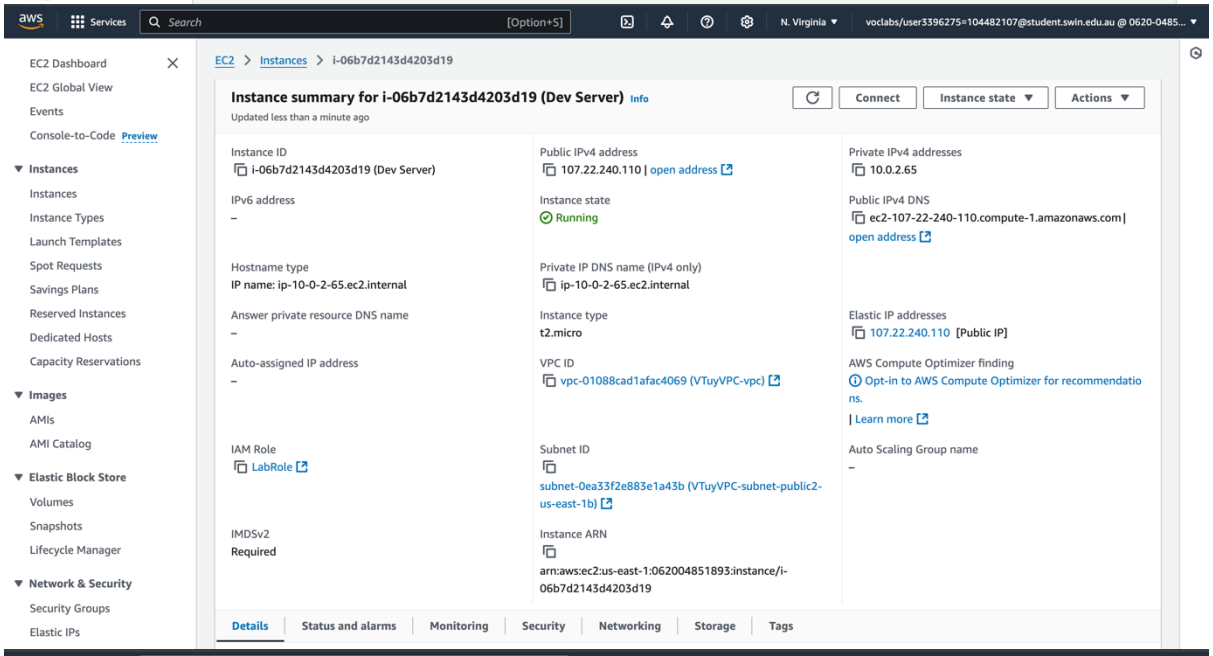
Addtionally, the auto scaling has been created and set up a target tracking scaling policy.

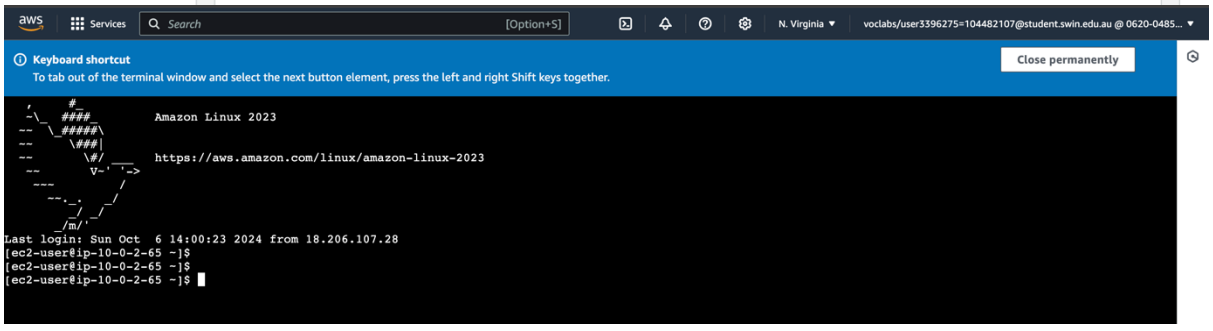


2.6 – EC2 instances

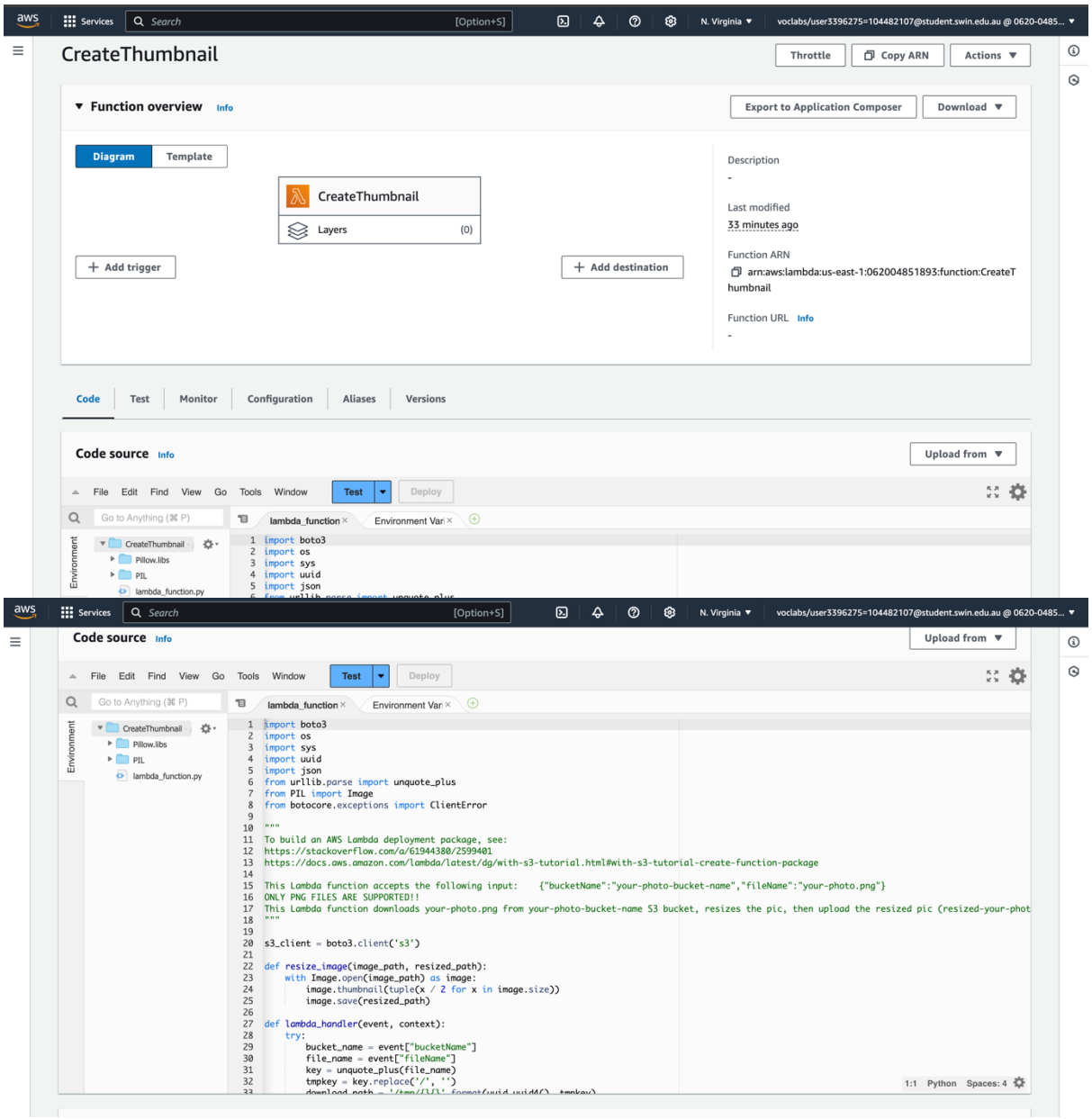
In addtion, I create Dev Server Instance adding 4 subnet(2 public and 2 private),by following the diagram in assignment 2.



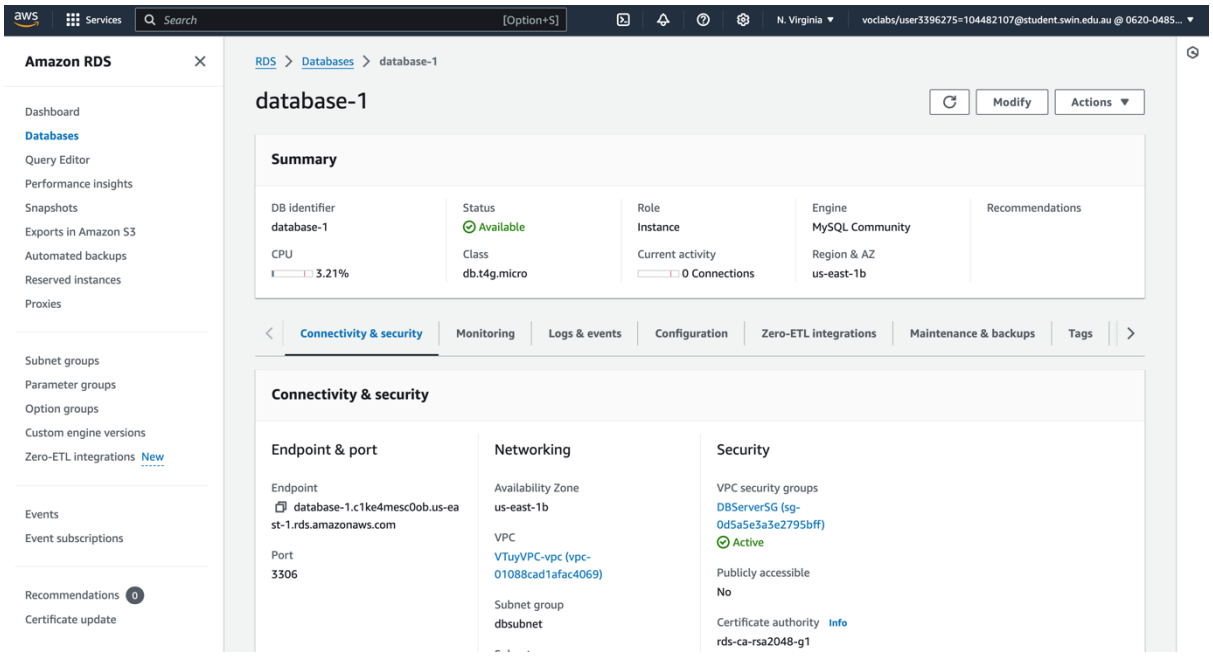




2.7 – CreateThumbnail Lambda function

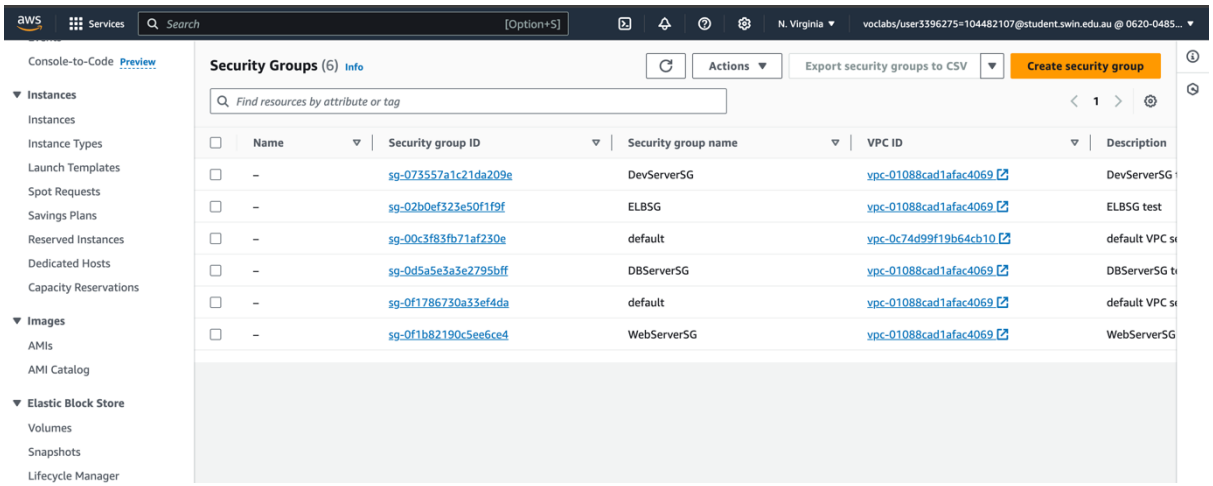


2.8 – Database with RDS



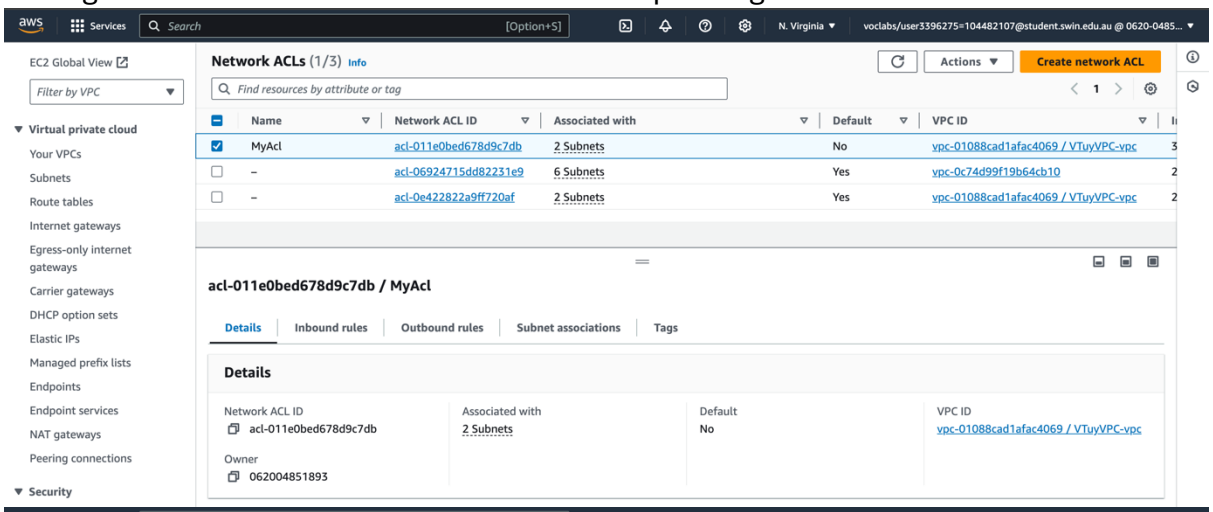
2.9 - Security groups

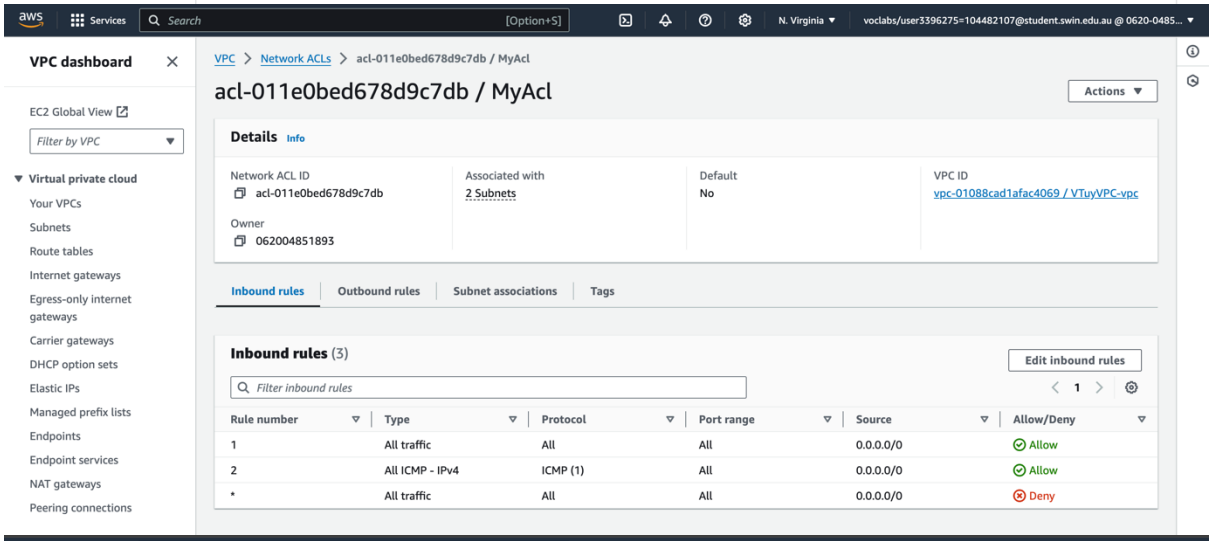
I have created 4 secuiry including DevServerSG, WebServerSG, ELVSG and DBServerSG which will be using for the load balancers and the DevServerSG will be used as the main SG.

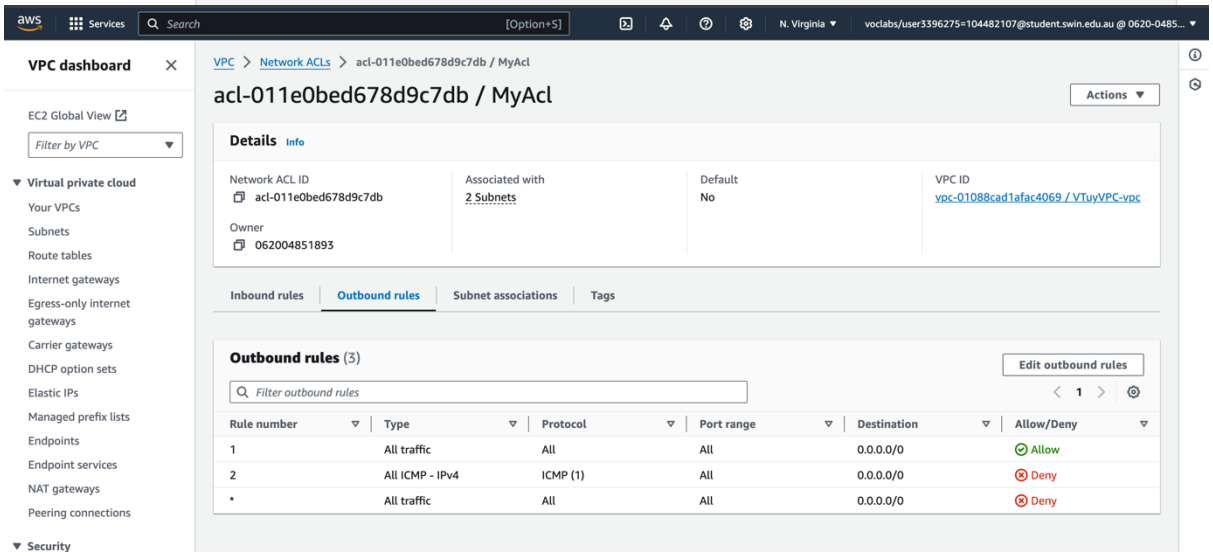


2.10 – Network ACL

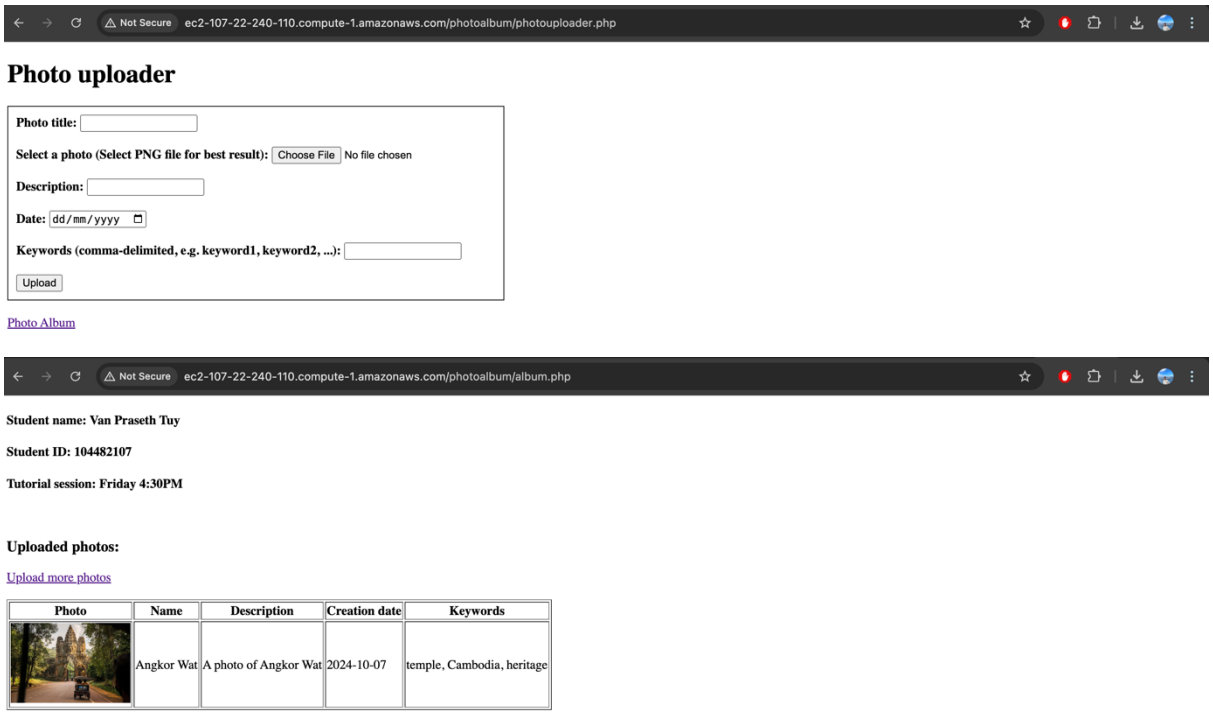
To enhance the security, I have addded an additonal layer of security to a network by setting limited access ICMP traffic to the corresponding subnets.







Testing



Link to photouploader.php:
http://ec2-107-22-240-110.compute-1.amazonaws.com/photoalbum/photouploader.php